Regression Diagnostics

POLS 602

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Review

1. What does LOESS let the data decide that a quadratic regression does not?
2. What does quantile regression estimate that OLS does not?
3. What are the Gauss-Markov assumptions?

Today

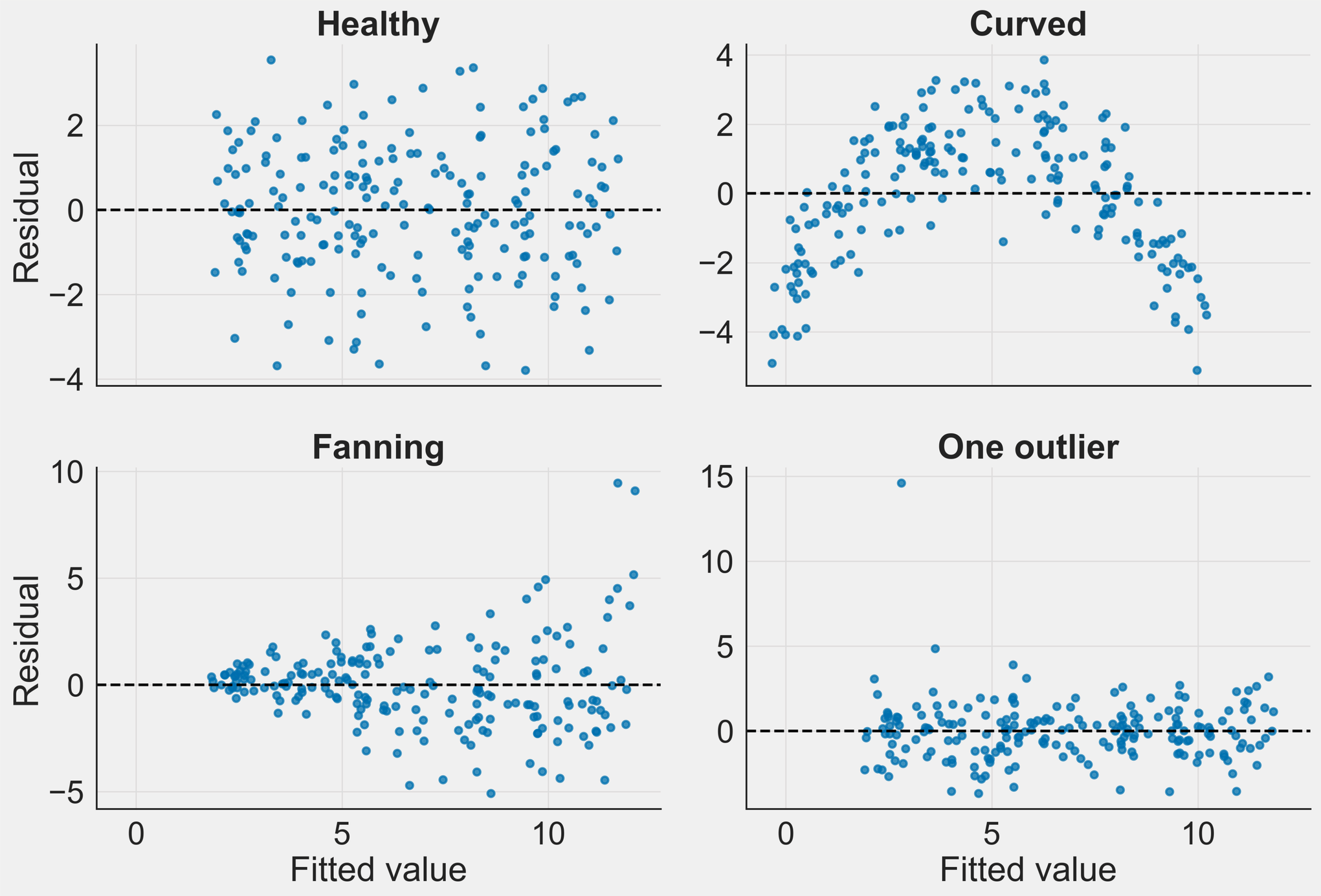
- How do we check whether the assumptions hold?
- What breaks when the errors are not normal?
- What breaks when the errors do not have constant variance?
- What breaks when the errors are correlated within groups?

Checking the Model

Assumptions Are About Errors

- The assumptions are about the errors ε_i
- We never observe the errors
- We observe the residuals $\hat{\varepsilon}_i = y_i - \hat{y}_i$
- Diagnostics look for structure in the residuals

Reading a Residual Plot



Three Problems, Three Questions

Problem	What to check
Non-normal residuals	Is the shape bell-like?
Heteroscedasticity	Is the spread constant?
Clustered errors	Do the errors move together within groups?

Non-Normal Residuals

The Problem

- Normality assumes $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$
- Residuals may be skewed, heavy-tailed, or lumpy
- *Do the residuals look like a draw from a bell curve?*

Why It Matters

- Unbiasedness and BLUE do not need normality
- Exact t and F tests do
- In large samples the CLT makes the tests approximately right
- If you want intervals on a single prediction though, the CLT doesn't help you

What Causes It

- A skewed outcome, such as income, wages, or counts
- Nonlinearity or omitted variables that the model leaves in the error
- Outliers or data errors
- Mixed groups with different means
- Bounded or binary outcomes

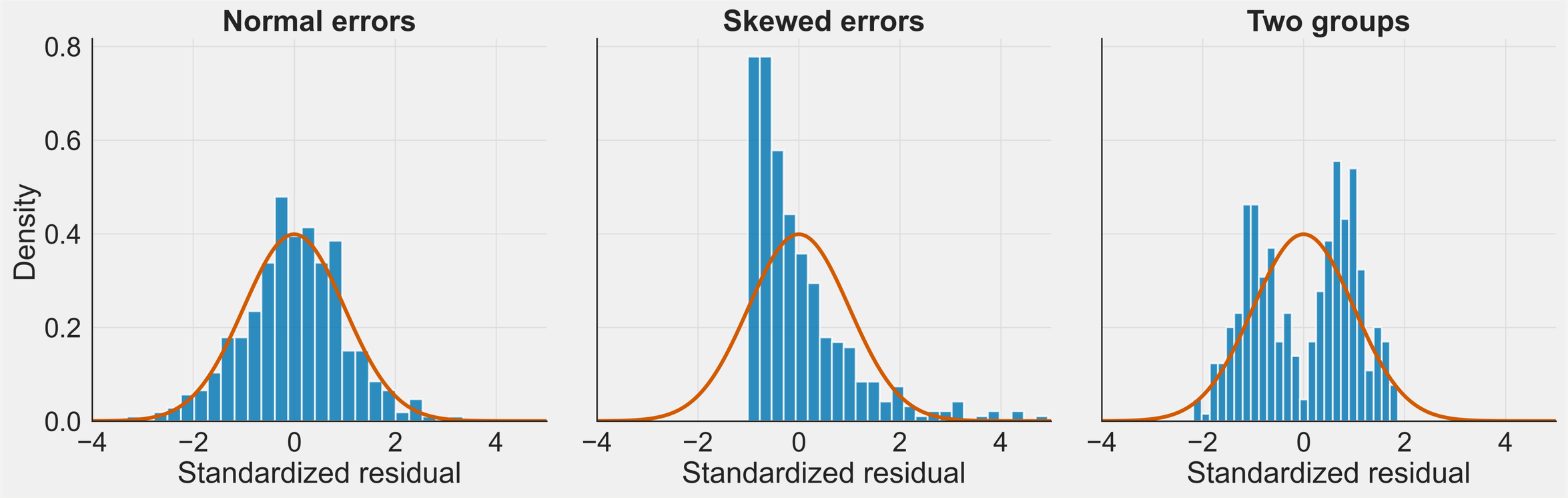
What It Implies

- Non-normality is usually a symptom, not the disease
- Check the functional form and the outcome’s scale first
- Check for outliers and mixed groups second
- Do not fix the residuals without asking why they look this way

Residual Histogram

- Plot a histogram of the residuals
- Overlay a normal curve with the same mean and variance
- Look at the shape: skew, tails, humps

Histograms Show Skew and Humps



What Is a Quantile?

- **Quantile**: the value q_τ below which a share τ of the distribution falls
- The 0.5 quantile is the median, and 0.25 and 0.75 are the quartiles
- Percentiles are quantiles written as percents
- *What value has 97.5% of a standard normal distribution below it?*

$$P(Z \leq q_\tau) = \tau, \qquad q_{0.975} = 1.96$$

What Is a QQ Plot?

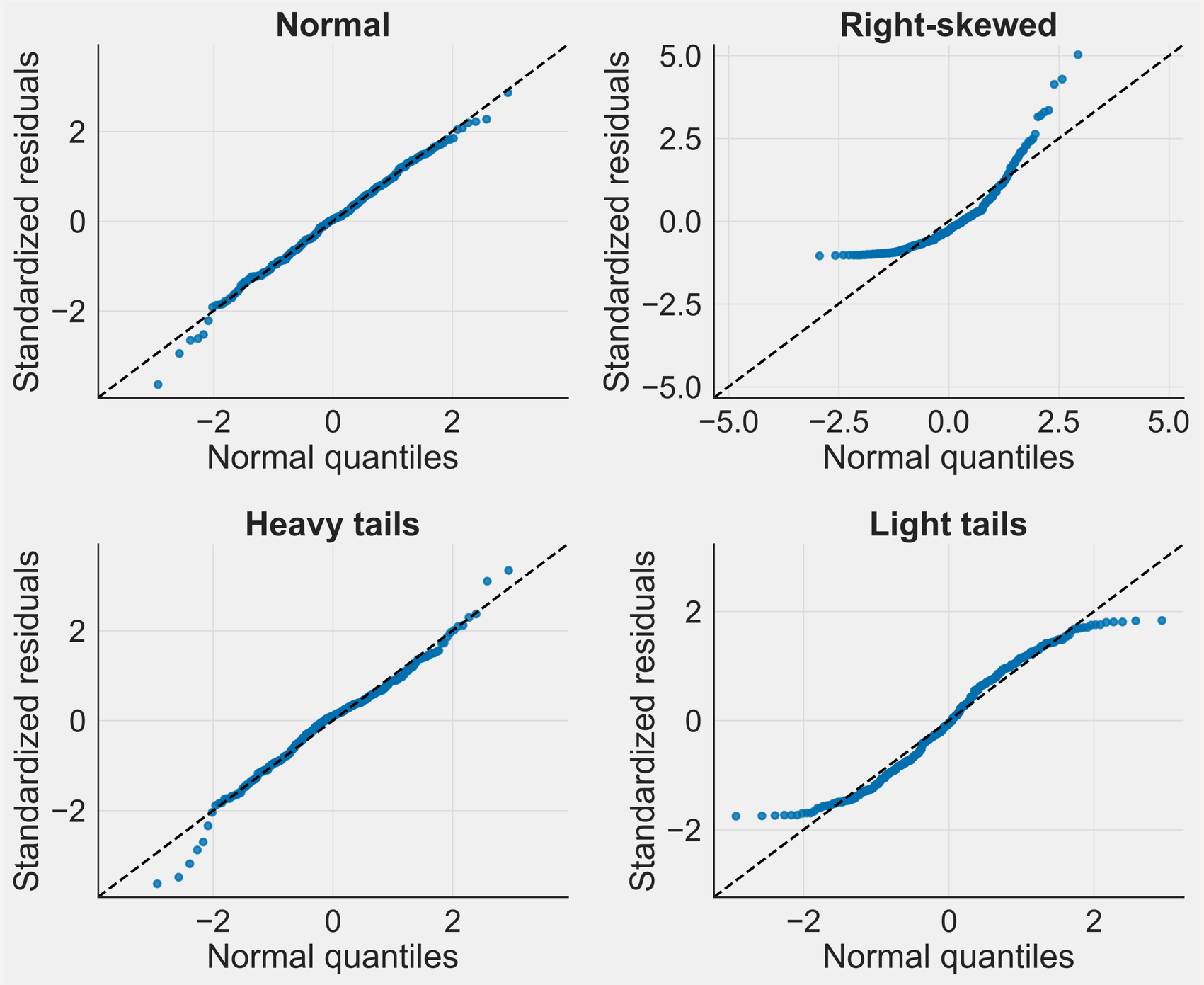
- **QQ plot** (quantile-quantile plot): the quantiles of the residuals plotted against the quantiles of a normal distribution
- If the residuals are normal, the points fall on a straight line
- Where the points leave the line shows how the shape differs from normal

- *Do the residuals have the quantiles that a normal distribution would give?*

How a QQ Plot Is Built

1. Standardize the residuals and sort them from smallest to largest
2. For each rank i, find the normal quantile at $(i - 0.5)/n$
3. Plot the sorted residuals against the normal quantiles
4. Add the 45-degree line

Reading a QQ Plot



Formal Tests of Normality

- A formal test asks whether the residuals are consistent with a normal distribution
- Null hypothesis: the residuals are normal
- A small p-value is evidence against normality
- Two common tests: Shapiro-Wilk and Jarque-Bera

The Shapiro-Wilk Test

$$W = \frac{\left(\sum_{i=1}^{n} a_i \, \hat{\varepsilon}_{(i)} \right)^2}{\sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2}}$$

- $\hat{\varepsilon}_{(i)}$: the ith smallest residual
- a_i: weights set by what a normal sample's ordered values look like
- W lies between 0 and 1, and W near 1 means normal
- Small W rejects normality

The Jarque-Bera Test

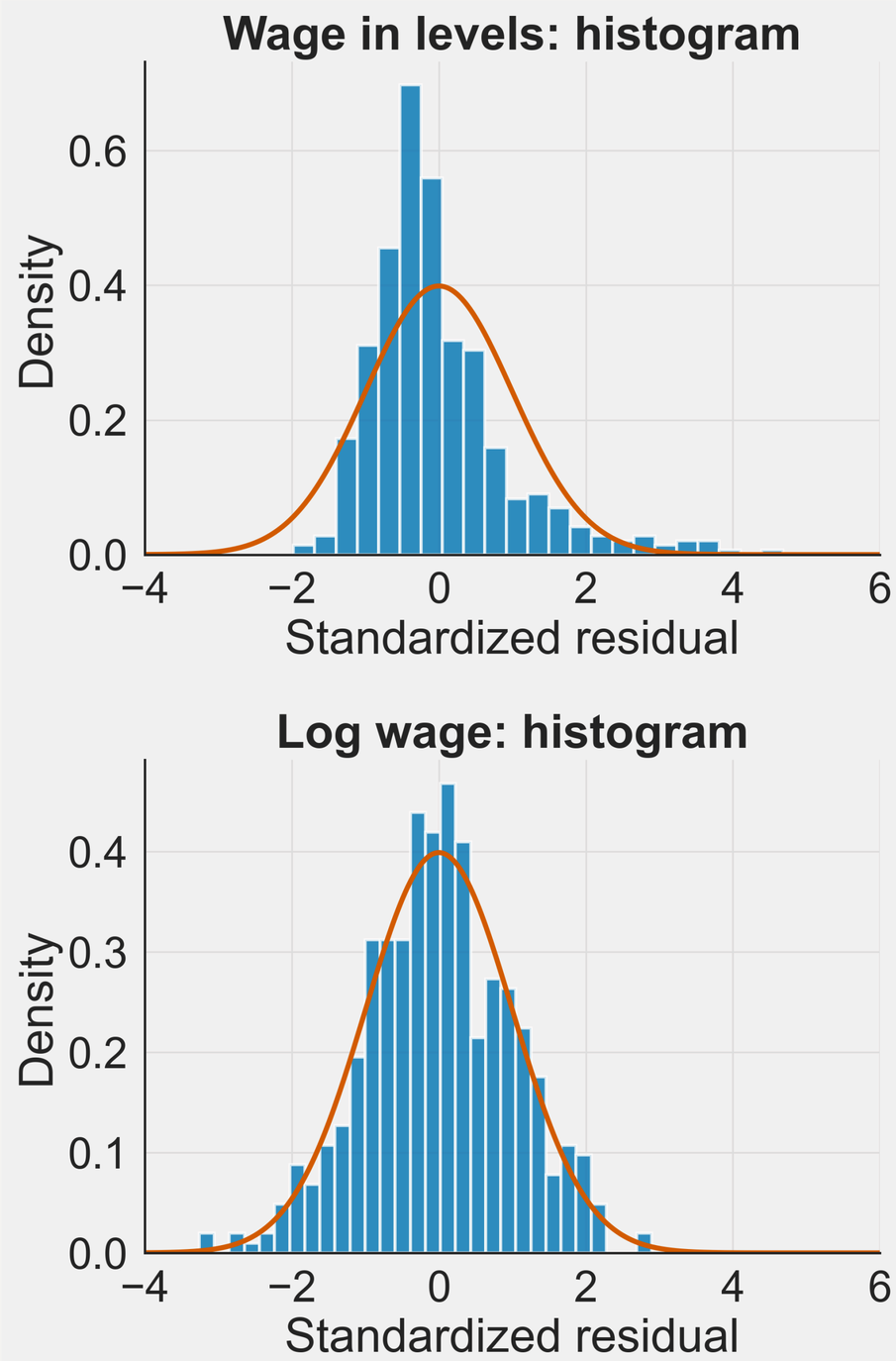
$$JB = \frac{n}{6}\left(g_1^2 + \frac{g_2^2}{4}\right)$$

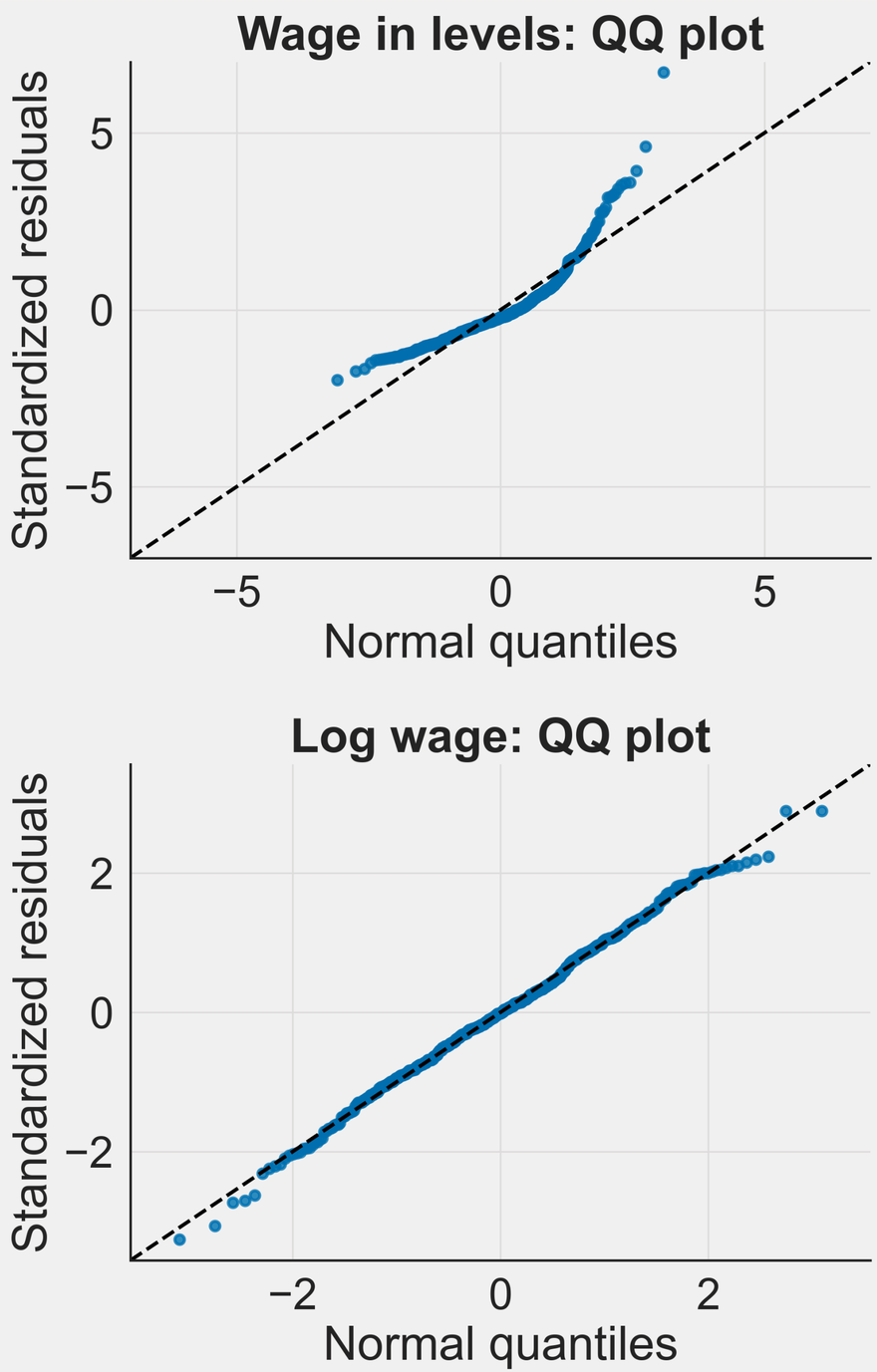
- JB is large when skewness (g_1) or excess kurtosis (g_2) is far from 0
- Reject at 5% if $JB > 5.99$

Cautions With Normality Tests

- Large samples reject for departures too small to matter
- Small samples have little power to detect real departures
- Failing to reject does not show the residuals are normal
- Use the tests alongside the plots, never in place of them

Skewed Outcome, Skewed Residuals





What to Do

- Find the cause before choosing a fix
- Transform a skewed outcome (logs)
- Repair the functional form or add missing groups
- Investigate outliers
- With a large sample and a focus on coefficients, rely on the CLT

Heteroscedasticity

The Problem

$$\text{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma_i^2 \quad \text{instead of} \quad \sigma^2$$

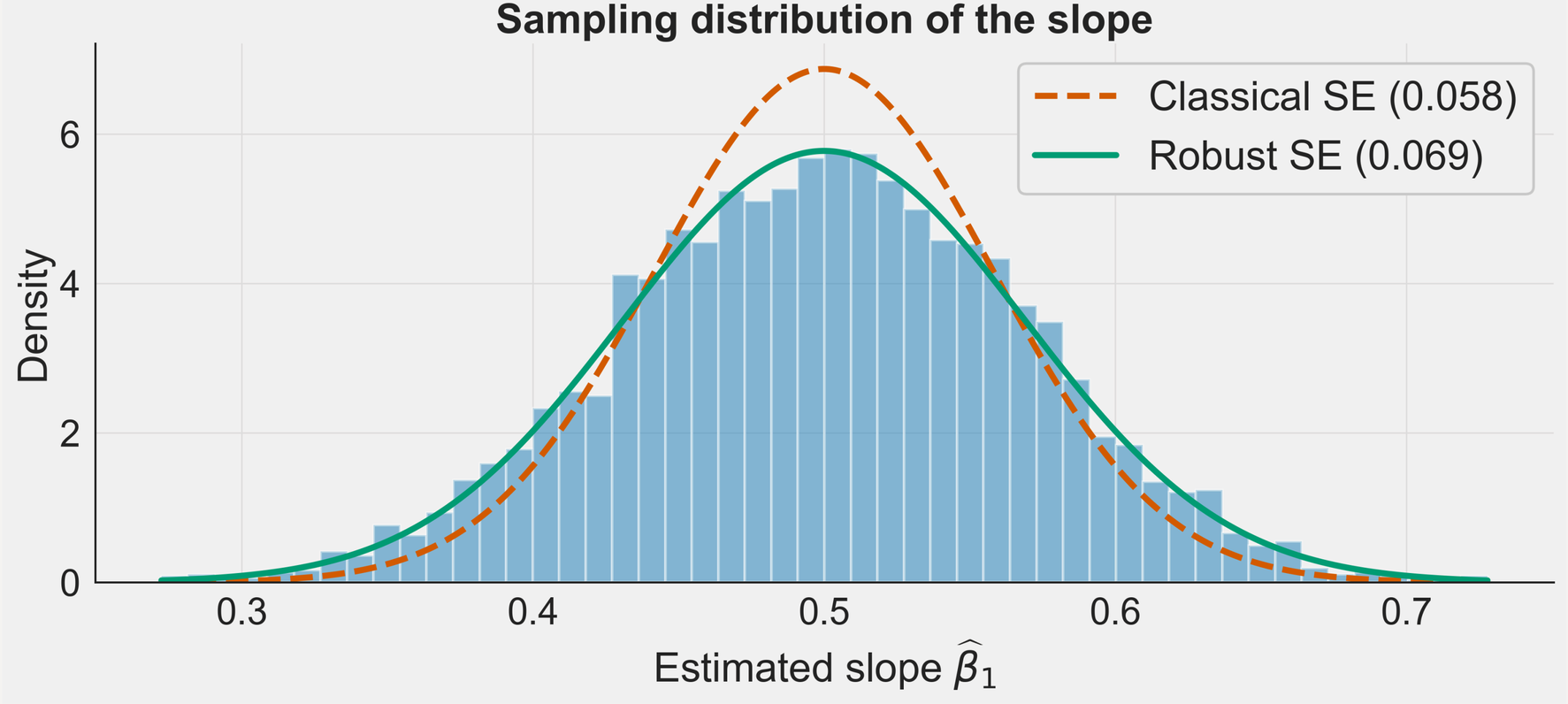
- The spread of the errors changes with $\mathbf{X}$
- *Are some observations noisier than others?*

Why It Matters

- OLS is still unbiased
- OLS is no longer the best linear unbiased estimator
- The usual standard errors are wrong, in either direction
- The usual t-tests, F-tests, and confidence intervals have the wrong error rates, even in large samples

The Standard Error Is Wrong

- 4,000 simulated samples, each with $n = 200$
- The true slope is 0.5, and the error variance grows with x
- Bars: the slope estimated in each sample
- Curves: the normal distribution implied by each standard error



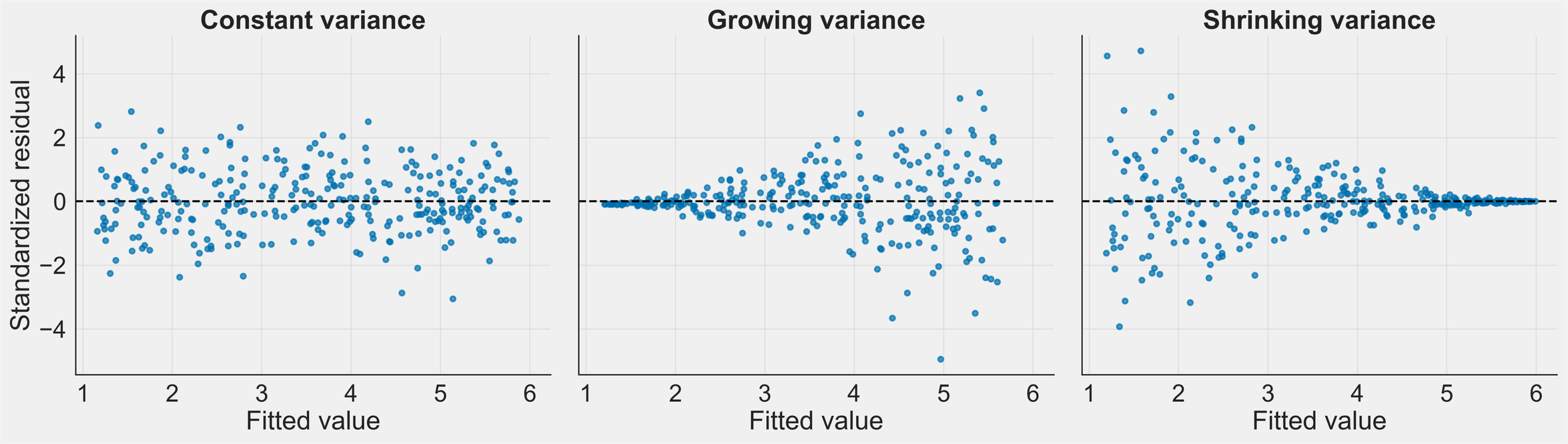
What Causes It

- The outcome’s scale: spending, income, populations
- Groups with different variability
- Averaged data with different group sizes
- Binary outcomes in a linear model
- Misspecification, such as an omitted variable or the wrong functional form

What It Implies

- Sometimes it is a feature of the data, and only the standard errors need fixing
- Example: attitudes on trade when one party’s voters are split into factions and the other’s are united
- Sometimes it is a symptom of misspecification
- Check the functional form and the outcome’s scale before patching the standard errors

Constant vs. Changing Variance



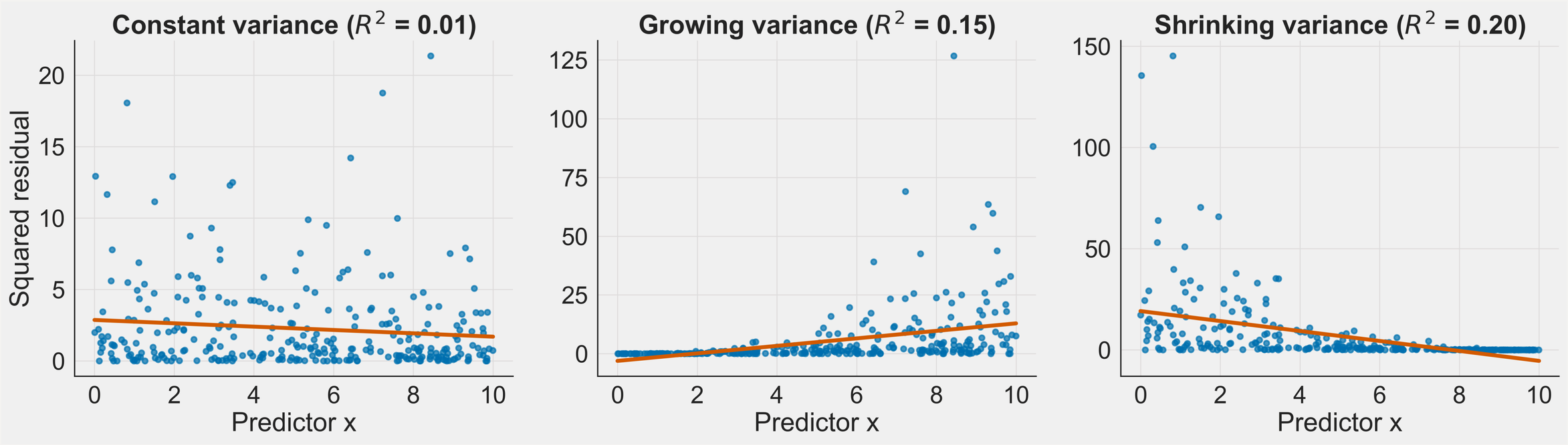
A Test for Changing Spread

- A residual plot is a judgment call
- A gradual change in spread is hard to see
- *Is the spread of the residuals related to the predictors?*
- **Breusch-Pagan test**: a test of the null that the error variance is constant

Squared Residuals Measure Spread

- The variance of an error is its expected squared value
- We see residuals, not errors, so $\widehat{\varepsilon}_i^2$ stands in for the variance at observation i
- Constant variance: the squared residuals are unrelated to the predictors
- Variance related to x: the squared residuals rise or fall with x

Squared Residuals Against x



The Breusch-Pagan Test

1. Fit the model and save the squared residuals $\hat{\varepsilon}_i^2$
2. Regress the squared residuals on the predictors
3. Compute $LM = nR^2$, where n is the sample size and R^2 comes from step 2
4. Compare LM to a χ^2 distribution with k degrees of freedom, where k is the number of predictors

$$\hat{\varepsilon}_i^2 = \delta_0 + \delta_1 x_{1i} + \cdots + \delta_k x_{ki} + u_i, \qquad H_0 : \delta_1 = \cdots = \delta_k = 0$$

Interpreting the Statistic

- Small LM: the predictors do not explain the squared residuals, so we fail to reject constant variance
- Large LM: they explain more than chance would, so we reject constant variance
- The direction of the relationship does not matter
- The cutoff is the χ^2_k critical value: 3.84 when $k = 1$ at the 5% level
- Failing to reject does not show that the variance is constant

Results for the Simulated Data

Case	R^2	$LM = nR^2$	p-value
Constant variance	0.012	3.6	0.059
Growing variance	0.155	46.4	< 0.001
Shrinking variance	0.202	60.6	< 0.001

- $n = 300$ and $k = 1$, so the 5% cutoff is 3.84
- It only detects variance that moves linearly with the predictors

Responding to Heteroscedasticity

- First, check the model: transform the outcome or add the missing structure
- If the unequal variance is a feature of the data, this creates two problems.

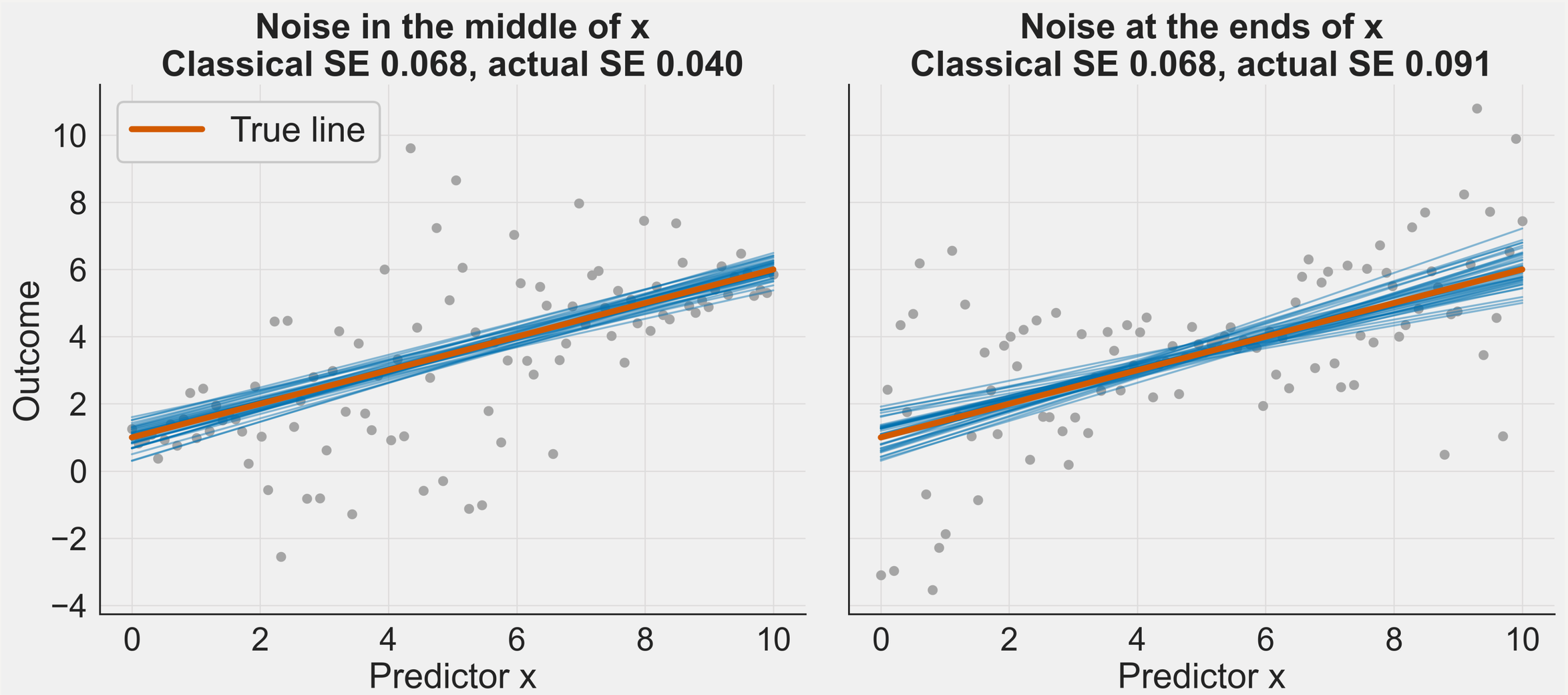
Problem	Fix	Changes the coefficients?
Standard errors are wrong	Robust standard errors	No
OLS is no longer the best estimator	Weighted least squares	Yes

Robust Standard Errors

Fixing the Standard Errors

- The OLS coefficients are still unbiased, so only the standard errors need fixing
- The usual formula pools all the residuals into one noise level, $\widehat{\sigma}^2$
- Pooling treats every observation as equally noisy
- *Does it matter which observations are noisy?*

Where the Noise Sits Matters



What Robust Standard Errors Do

- Keep the OLS coefficients
- Relax the assumption of a constant variance
- Measure each observation’s noise with its own squared residual
- Count that noise more for observations far from $\bar{x}$

The Sandwich Formula

The variance of the OLS coefficients, with no assumption about the errors:

$$\text{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \; \mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} \; (\mathbf{X}^\top \mathbf{X})^{-1}$$

- Bread: depends only on $\mathbf{X}$
- Meat: where the errors enter
- $\boldsymbol{\Omega} = \text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X})$: an $n \times n$ matrix with the variance of each error on the diagonal and the covariance of each pair off it

What Is in $\boldsymbol{\Omega}$?

Under homoscedasticity with uncorrelated errors:

$$\boldsymbol{\Omega} = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}$$

- Diagonal: $\operatorname{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma^2$, the same for every observation
- Off the diagonal: $\operatorname{Cov}(\varepsilon_i, \varepsilon_j \mid \mathbf{X}) = 0$

Under Homoskedasticity

Substitute $\boldsymbol{\Omega} = \sigma^2 \mathbf{I}$. Since σ^2 is a number, it moves to the front:

$$(\mathbf{X}^\top \mathbf{X})^{-1}\ \mathbf{X}^\top \sigma^2 \mathbf{I}\, \mathbf{X}\, (\mathbf{X}^\top \mathbf{X})^{-1}\ = \sigma^2\, (\mathbf{X}^\top \mathbf{X})^{-1}\ \mathbf{X}^\top \mathbf{X}\, (\mathbf{X}^\top \mathbf{X})^{-1}$$

The meat cancels one slice of bread:

$$\mathrm{Var}(\,\widehat{\boldsymbol{\beta}}\ \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

- This only works because one σ^2 can be pulled out

Allowing Different Variances

Now give each observation its own variance:

$$\boldsymbol{\Omega} = \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n^2 \end{bmatrix}$$

No single σ^2 can be pulled out, so the meat is a sum over the rows of $\mathbf{X}$:

$$\mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} = \sum_{i=1}^{n} \sigma_i^2 \, \mathbf{x}_i \mathbf{x}_i^\top$$

The Sandwich Estimator

- We cannot estimate each σ_i^2, since we see one draw per observation
- So we use $\hat{\varepsilon}_i^{\,2}$ in place of σ_i^2

$$\widehat{\mathrm{Var}}\left(\widehat{\boldsymbol{\beta}}\right) = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2}\, \mathbf{x}_i \mathbf{x}_i^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1}$$

- Each term is noisy, but the noise averages out across the sum
- The result is a $(k+1) \times (k+1)$ matrix with each coefficient's estimated variance on the diagonal
- **Robust standard errors**: the square roots of those variances

Versions of Robust Standard Errors

Version	Replace $\hat{\varepsilon}_i^2$ with	Use for
HC0	$\hat{\varepsilon}_i^2$	Large samples
HC1	$\hat{\varepsilon}_i^2 \cdot n/(n-k-1)$	Stata
HC2	$\hat{\varepsilon}_i^2/(1-h_{ii})$	Experiments
HC3	$\hat{\varepsilon}_i^2/(1-h_{ii})^2$	Samples < ~250

- Residuals run smaller than errors, especially for unusual x
- h_{ii}: leverage, how unusual $\mathbf{x}_i$ is (next lecture)
- They agree in large samples, but software defaults differ

Robust Standard Errors in Practice

- The coefficients do not change
- The standard errors can go up or down
- Valid without knowing the form of the heteroscedasticity
- They do not fix a wrong regression equation, such as one that omits a predictor
- The cost is some loss of precision compared with a correctly weighted estimator

Clustered Errors

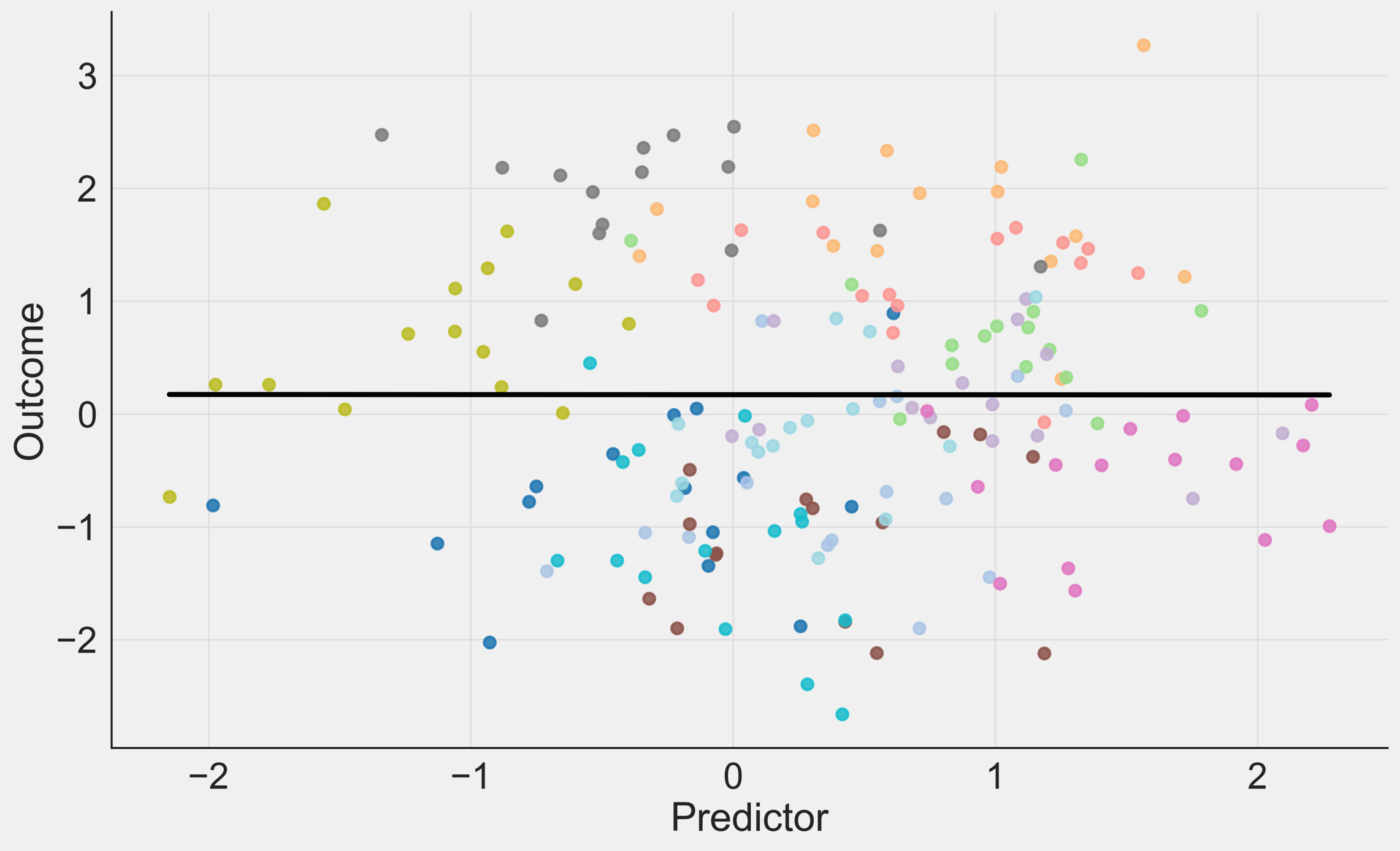
The Problem

- **Cluster**: a group of observations that share something we did not model
- Examples: students in a school, legislators in a state, repeated observations of one unit
- Errors within a cluster are correlated
- *Does a second student from the same school tell us as much as a student from a new school?*

Why It Matters

- OLS is still unbiased if the shared shocks are unrelated to the predictors
- The usual standard errors treat the observations as independent
- Correlated observations carry less information than independent ones
- The standard errors are too small

Shared Shocks Within Clusters



Cluster-Robust Standard Errors

$$\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{g=1}^{G} \mathbf{X}_g^\top \widehat{\boldsymbol{\varepsilon}}_g \widehat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}$$

- $g = 1, \ldots, G$ indexes the clusters
- $\mathbf{X}_g$ and $\widehat{\boldsymbol{\varepsilon}}_g$: the rows of $\mathbf{X}$ and the residuals in cluster g
- Same bread, new meat: the sum runs over clusters, not observations

Choosing the Clusters

- Cluster at the level where errors are correlated
- Cluster at the level of sampling or treatment assignment
- Need many clusters, roughly 30 to 50 or more
- The meat has one term per cluster, so G acts as the sample size, not n
- With few clusters, standard errors are underestimated

Weighted Least Squares

Why Change the Estimator?

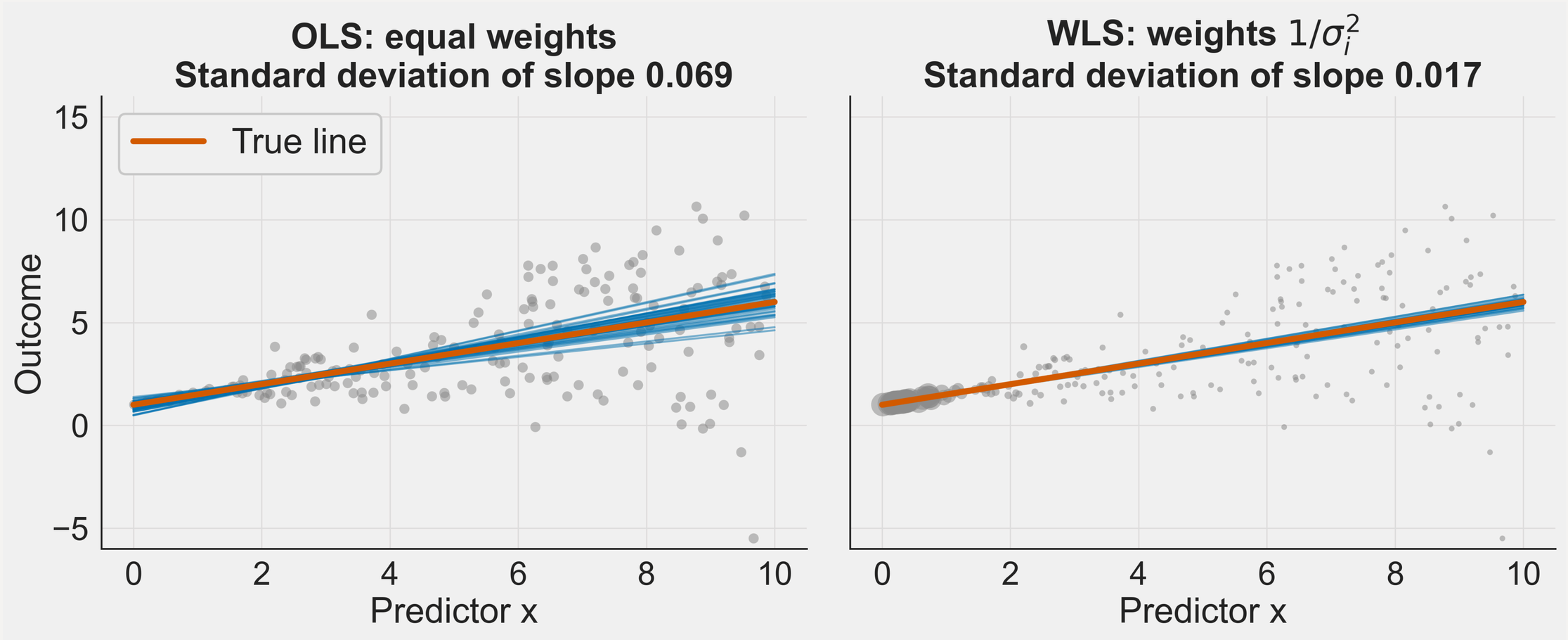
- Robust standard errors correct the standard errors, but the coefficients are still the OLS estimates
- OLS gives every observation the same weight, whatever its error variance
- Under heteroscedasticity, OLS is unbiased but not BLUE: another linear unbiased estimator has a smaller variance
- Example: a state average from 5,000 respondents has 1/100 the variance of one from 50
- *Should those two observations have the same weight?*

What WLS Does

- **Weighted least squares (WLS)**: OLS with a weight on each observation
- Precise observations, those with a small error variance, receive large weights
- Imprecise observations, those with a large error variance, receive small weights
- The fitted line therefore stays closer to the precise observations
- The coefficients change, and they vary less from sample to sample

WLS Reduces Sampling Variability

- Each blue line is one sample's fit; larger points have more weight
- Both center on the true line, but the WLS lines vary far less



The WLS Objective

WLS multiplies each squared residual by a weight w_i:

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- The weight is the inverse of the error variance: $w_i = 1/\sigma_i^2$
- A large w_i makes a large residual at observation i contribute more to the sum of squared errors. Thus, the line of best fit will be pulled towards that point.
- If every σ_i^2 is equal, every weight is equal and WLS is OLS

Why Inverse-Variance Weights?

Start with the model and divide every term by σ_i:

$$\frac{y_i}{\sigma_i} = \frac{\mathbf{x}_i^\top}{\sigma_i}\boldsymbol{\beta} + \frac{\varepsilon_i}{\sigma_i}$$

- $\boldsymbol{\beta}$ does not change, because both sides are divided by the same number
- The new error has the same variance for every observation:

$$\mathrm{Var}(\varepsilon_i/\sigma_i \mid \mathbf{X}) = \sigma_i^2/\sigma_i^2 = 1$$

- With uncorrelated errors, the rescaled model meets the Gauss-Markov assumptions, so OLS on it is BLUE

OLS on the Rescaled Model Is WLS

OLS on the rescaled model minimizes the sum of squared rescaled residuals:

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- The right side is the WLS objective, with $w_i = 1/\sigma_i^2$
- So WLS with these weights is BLUE

The WLS Estimator

Minimizing the WLS objective gives:

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- This is the OLS formula with $\mathbf{W}$ inserted; if $\mathbf{W} = \mathbf{I}$, it is OLS
- With $w_i = 1/\sigma_i^2$, $\mathbf{W} = \mathbf{\Omega}^{-1}$ and the variance simplifies:

$$(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{\Omega} \mathbf{W} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$$

$$\text{Var}(\widehat{\boldsymbol{\beta}}_{WLS} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$$

Generalized Least Squares

- WLS is a special case of generalized least squares
- **Generalized least squares (GLS)**: the extension of WLS to any known $\boldsymbol{\Omega}$, including correlated errors

$$\widehat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{Y}$$

- Both address heteroscedasticity, but WLS assumes observations are independent. GLS allows observations to be dependent (autocorrelation)
- Commonly used for time and spatial data

When the Weights Are Wrong

- Weights are wrong if they are not proportional to $1/\sigma_i^2$, as when the variance model is wrong
- WLS is still unbiased if the mean model is correct, meaning $E[\varepsilon_i \mid \mathbf{X}] = 0$
- But it is not BLUE, and its variance can be larger than the variance of OLS
- Also, $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ is the wrong variance, so the usual WLS standard errors are wrong
- Robust standard errors for WLS correct this

Linear Models Are Approximations

- Unbiasedness and BLUE assume that $E[y_i \mid \mathbf{X}]$ is exactly linear, which is rarely true
- Otherwise, each estimator gives a linear approximation to the true mean
- OLS counts all observations equally; WLS fits best where the weights are large
- Example: if the true mean is $6\sqrt{x_i}$ and the variance grows with x_i, the OLS slope approaches 1.5 and the WLS slope approaches 2.3
- So a large gap between OLS and WLS estimates is a reason to check the mean model

Robust SEs or WLS?

- OLS with robust SEs is the default, because it requires no variance model
- Its cost is less precision than WLS with correct weights
- Neither corrects a wrong mean model, such as an omitted predictor

Putting It Together

Diagnostics at a Glance

Problem	Tools	Responses
Non-normal residuals	Histogram, QQ plot	Transform, fix the model
Heteroscedasticity	Residual plot, Breusch-Pagan	Robust SEs, WLS
Clustering	Study design	Clustered SEs

Summary

- Diagnostics look for structure in the residuals, which points to a failed assumption
- Non-normal residuals do not bias OLS, but they often signal a misspecified model
- Heteroscedasticity leaves OLS unbiased but makes the usual standard errors wrong
- Clustered errors leave OLS unbiased but make the usual standard errors too small

Key Definitions

Key Definitions

- **Residual plot**: residuals against fitted values, used to spot curvature, changing spread, and unusual points
- **Prediction interval**: an interval for a single new outcome, which depends on the error distribution
- **Quantile**: the value q_τ below which a share τ of the distribution falls
- **QQ plot**: residual quantiles against normal quantiles; points on the line mean normality

Key Definitions

- **Shapiro-Wilk test**: a test of the null that the residuals are normal; W near 1 means normal
- **Skewness**: asymmetry of a distribution; 0 for a normal
- **Excess kurtosis**: tail weight relative to a normal; 0 for a normal
- **Jarque-Bera test**: a test of normality based on skewness and excess kurtosis; $JB \approx \chi^2_2$ under the null

Key Definitions

- **Heteroscedasticity**: error variance that differs across observations
- **Breusch-Pagan test**: a test of constant variance; regress the squared residuals on the predictors
- **LM statistic**: $LM = nR^2$ from that regression; approximately χ^2_k under the null
- **Robust standard errors**: standard errors that stay valid under heteroscedasticity (the sandwich estimator)
- **Sandwich estimator**: *[Math Processing Error]*

Key Definitions

- **Cluster**: a group of observations that share something we did not model
- **Clustered errors**: errors correlated within groups of observations
- **Cluster-robust standard errors**: a sandwich estimator that sums over clusters instead of observations

Key Definitions

- **Weighted least squares**: minimizes $\sum w_i (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$ with $w_i = 1/\sigma_i^2$
- **Generalized least squares**: the extension of WLS to any known $\boldsymbol{\Omega}$, including correlated errors